AtomRemote Agent7 - Controlled Phased Roadmap

Project root: E:\AtomosRemoteWithBackend | Native: native-android | Backend: backend | Cloud relay only after local hardware certification

Decision after 20.01.0 logs: We can continue, but the next phase must fix/diagnose AppPolling transaction arbitration and typed payload correlation before expanding Zoom, Grid, Codec, Audio, Output or other inner settings.

Current verified baseline

Area	Status	Evidence / interpretation
TCS pairing	FROZEN	ID_SLOTS -> CONNECT_NAME -> CCID -> BCID -> accept -> ~CCID -> ACCEPTED
PacketLink	WORKING / FROZEN	READY/WRITING/READING transport retained; do not redesign
Record / Stop / Play / Pause	WORKING / FROZEN	Existing controls must not regress
Backend agent / FIFO / reconnect	WORKING	Local backend registration and command path available
Input 3/3	HARDWARE PASS	Source, HDR State, Device Name return real recorder-backed payloads
State page polling 0x0640-0x0649	NOT PASS	receivedPages=0; NO_TYPED_PAGE_PAYLOAD; PACKETLINK_DELIVERED_ONLY

What the 20.01.0 logs tell us

The first test run hit the backend before PacketLink was ready and correctly returned 502/blocking errors. The second run, after PacketLink was ready, restored the real Input collector and produced 3/3 responses. That proves the backend-agent-PacketLink route is still usable.

However, the state endpoints still returned cycleComplete with receivedPages=0 and empty pages. Android logs show the bootstrap issuing many RemoteApi queries, receiving E0 transport frames, and explicitly logging "PacketLink transport complete; recorder action unconfirmed". This means the state-page layer is not yet validated as recorder state. It may be polling arbitration, wrong GET/response correlation, wrong page semantics, or a combination.

Therefore: do not jump directly to UI feature expansion. Phase 20.02 must establish a clean single-owner transaction window and capture the actual RemoteApi payload path. After that gate passes, we proceed tab by tab.

Global phase rules - apply to every Agent7 phase

Rule	Required behavior
Backup before patch	Create E:\AtomosRemoteWithBackend\agent7-backups\AtomRemote-Agent7-_ before touching source.
Install	Version-specific INSTALL script with preflight guards and explicit changed-file manifest.
Verify	VERIFY script must validate expected source markers and frozen-code invariants.
Build	Backend typecheck/build + Android assembleDebug must pass.
Install APK	Optional -AdbSerial parameter installs APK and reapplies adb reverse tcp:8787.
Hardware test	Every phase includes a dedicated TEST-HARDWARE script.
Log capture	Every phase includes TEST-AND-CAPTURE plus a standalone CAPTURE-LOGS script.
Result policy	HTTP 200, TX OK, queued, or E0 alone is NOT PASS.
PASS gate	PASS requires recorder payload/readback and, where applicable, visible recorder confirmation.
Restore	Each phase includes a restore path using the phase backup.
Frozen zones	No changes to TCS pairing, PacketLink architecture, Record/Stop/Play/Pause unless separately approved.

Required package layout

```
E:\agent7\AtomRemote-Agent7-<version>-<name>\
INSTALL-AGENT7-<version>.ps1
VERIFY-AGENT7-<version>.ps1
BUILD-INSTALL-AGENT7-<version>.ps1
TEST-AGENT7-<version>-HARDWARE.ps1
TEST-AND-CAPTURE-AGENT7-<version>.ps1
CAPTURE-AGENT7-<version>-LOGS.ps1
RESTORE-AGENT7-<version>.ps1
PATCH-MANIFEST.txt
docs\
files\
```

Phase 20.02.0 - Polling Arbitration + Typed Payload Correlation

Work: Make explicit GET/SET transactions own PacketLink until their response is resolved; AppPolling must yield and resume afterward. Instrument packetId, command ID, expected response ID, m_isLocal, TX atom, RX payload, consumer, timeout, and completion reason. Do not delete polling. Do not alter TCS or PacketLink framing.

Exit gate: A controlled explicit GET completes without a polling command stealing its response; logs show one owner, correlated response, and actual payload candidate or a precise protocol-level reason why none exists.

Phase 20.03.0 - RemoteApi Protocol Matrix + Input Complete

Work: Use Contents.zip and current native source to build the authoritative command matrix. Finish Input inner settings that are actually supported: source, HDR/gamma, gamut/camera metadata, current signal/format, device identity, HDMI/SDI-related values where proven.

Exit gate: Every Input row has exact GET/SET IDs, response ID, protobuf schema, m_isLocal, packetId behavior, supported range/list, backend route, Android coordinator, UI target, and hardware readback.

Phase 20.04.0 - Monitor Core + Zoom

Work: Wire Monitor mode state and complete Zoom using recovered Zoom GET/SET/list/box/pan commands. Preserve already-working monitor behavior.

Exit gate: Zoom supported list is recorder-backed; SET changes AG2; subsequent GET returns the new value; UI shows confirmed state only.

Phase 20.05.0 - Monitor Assist Complete

Work: Wire Focus, Zebra, False Color, Blue Only and all confirmed inner settings. False Color must be treated as a parent feature with child settings, not only an On/Off switch. Recover thresholds, modes, presets, scales or ranges only where Contents/AG2 proves them.

Exit gate: Each child setting has recorder-backed GET/SET/readback. Unsupported children are hidden or marked unsupported, never guessed.

Phase 20.06.0 - Scopes

Work: Implement waveform/vectorscope/parade/histogram and size/position/opacity/mode only where RemoteApi and AG2 support them. Validate existing 0x0258/0x025A/0x025C paths against real responses.

Exit gate: Scope state is read from recorder; changing a supported parameter causes visible recorder change and readback.

Phase 20.07.0 - Guides + Grid + Safe Area

Work: Repair Frame Guide, 3x3/9-grid, guide format/aspect, Safe Area and related child settings. Recover full protobuf fields so changing one field does not clobber siblings. Investigate double-frame Safe Area behavior.

Exit gate: Grid and Safe Area can be changed independently; GET reflects exact active values; no double-frame regression; visible AG2 confirmation.

Phase 20.08.0 - Record + Codec

Work: Keep Record Start/Stop transport frozen. Repair Codec using exact RecordCodecProfile protobuf, supported profiles, bit depth, restart/non-reboot semantics and readback. Add only confirmed Record inner settings.

Exit gate: Codec SET is visibly applied on AG2 and subsequent GET returns the selected codec/bit depth. Unsupported values are rejected before TX.

Phase 20.09.0 - Audio

Work: Recover channel list, source, record enable, monitor enable, gain, mute/pairing, headphone and audio-meter data where supported. Separate meter telemetry from configurable settings.

Exit gate: Per-channel state and configurable values are recorder-backed. Audio meters are not considered configuration readback.

Phase 20.10.0 - Output

Work: Wire HDMI/SDI output source, format, HDR type, gamut, legalize/down-convert/LUT options and standards only when supported.

Exit gate: Output values come from recorder, safe SETs visibly change output/device UI where observable, and GET confirms.

Phase 20.11.0 - Playback

Work: Preserve Play/Pause. Add playback state, clip/seek/skip/loop/speed/frame navigation/format only when confirmed by RemoteApi.

Exit gate: Transport controls remain regression-free; added settings have exact correlated readback.

Phase 20.12.0 - Device / Settings

Work: Wire device name/model/firmware/AirGlu identity plus brightness, display, fan, date/time, language, storage/system settings where supported. Avoid destructive operations by default.

Exit gate: Identity is real recorder data; safe settings round-trip with readback. Destructive settings require manual confirmation.

Phase 20.13.0 - Timecode + Sync

Work: Wire TC source, mode, drop/non-drop, record/free run, jam/sync, AirGlu sync, frame rate, current TC and offset where supported.

Exit gate: Timecode values remain stable, do not break connected lifecycle, and readback matches recorder.

Phase 20.14.0 - Media

Work: Wire media presence, type, capacity, remaining time, target/slot/status and health where supported. Format-media remains guarded/manual.

Exit gate: Read-only media state is recorder-backed. Destructive operations are never auto-tested.

Phase 20.15.0 - Other Confirmed RemoteApi Families

Work: Add only additional tabs/settings proven by Contents.zip and AG2 capability behavior. Distinguish RemoteApi-supported from AG2-supported.

Exit gate: No fabricated tabs. Each row is classified SUPPORTED, UNSUPPORTED_BY_AG2, BLOCKED, or PASS with evidence.

Phase 20.16.0 - Android UI Integration

Work: Bind all validated backend states to the UI. Show current value, supported values, pending state, confirmed state, reject/error state, and last readback. Inner settings screens mirror real feature hierarchy.

Exit gate: UI never permanently changes on TX alone; it updates after confirmed recorder readback.

Phase 20.17.0 - Local End-to-End Certification

Work: Run full local backend + wireless ADB + agent + FIFO + reconnect + all GETs + safe SETs + app restart/backend restart/Bluetooth reconnect regression suite.

Exit gate: All applicable AtomRemote Sheet rows are PASS/UNSUPPORTED/BLOCKED with evidence; no frozen-feature regression.

Phase 20.18.0 - Cloud Relay / Vercel

Work: Only after local certification. Wire authenticated cloud relay while preserving local mode, FIFO, request IDs, idempotency, timeout/reconnect and result propagation.

Exit gate: Remote request produces the same hardware-confirmed result semantics as local mode; no cloud layer may convert TX success into hardware PASS.

Master per-setting checklist

Field	Must be completed before row can be closed
Feature hierarchy	Tab -> parent setting -> inner/child setting
Command evidence	Exact command ID and exact response ID
Payload	Protobuf message/schema, fields, enum/range/list
Transport flags	m_isLocal and packetId rule
Read path	Recorder payload parser + state model
Write path	SET encoder + expected response/readback
Supported values	Recorder-backed list/range or proven static enum
Backend	Action registry + API route + timeout/error mapping
Android	Coordinator/FIFO ownership + correlation
UI	Current, pending, confirmed, rejected states
Hardware test	PowerShell test + visible/readback confirmation
Evidence	Contents.zip / Android log / recorder payload / visible AG2 state
Final status	PASS / FAIL / BLOCKED / UNSUPPORTED / MANUAL_CONFIRMATION

Standard test sequence for every phase

1. Install phase and create backup.
2. Verify frozen source markers and changed-file markers.
3. Backend typecheck/build.
4. Android build + APK install + adb reverse.
5. Connect/pair AG2 normally and wait for PacketLink READY.
6. Clear targeted logcat.
7. Run phase hardware GET test.
8. Run safe SET test only when explicitly included by the phase.
9. Immediately perform recorder GET/readback.
10. Capture filtered Android logs and backend output.
11. Write SUMMARY.csv and SUMMARY.txt.
12. Advance only if exit gate is satisfied; otherwise patch within the same phase version family.

20.02.0 exact focus - next phase

The next patch should not attempt Zoom, Grid, Codec, False Color inner settings or other tabs yet. It should instrument and arbitrate the state/feature transaction path so that an explicit request has exclusive ownership until its correlated response is handled. AppPolling may resume afterward. The goal is to determine whether the missing typed payload is caused by polling interference, response correlation, page command semantics, m_isLocal/packetId behavior, or parser logic.

20.02.0 GO condition: current frozen functionality remains intact; Input 3/3 still passes; at least one controlled explicit RemoteApi GET has a fully traceable TX -> RX -> consumer -> correlation result without AppPolling interleaving. If the recorder truly returns only E0, the phase must produce enough evidence to identify the next protocol correction rather than falsely passing.

Decision

YES - we can proceed. The logs do not show a reason to roll back 20.01.0. Input 3/3 is again working after PacketLink becomes ready, and the frozen PacketLink transport is still moving commands. But the next safe step is Phase 20.02.0, not feature expansion. State-page results are still transport-only and must be made trustworthy first.